Present Day, Present Time: AI Exposure and Labor Market Risk

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Building on Trade Theory

- **Task-based production:** firms combine a *continuum of tasks*, not just labor and capital
  - Tasks differ in how easily they can be delegated to a cheaper alternative
- **Grossman & Rossi-Hansberg (2008) — "Trading Tasks"**
  - Offshoring = delegating tasks to cheaper foreign labor
  - A falling offshoring cost $\rightarrow$ more tasks sent abroad $\rightarrow$ wage pressure on affected workers
- **Ottaviano, Peri & Wright (2013)**
  - Workers differ in type i^*; different groups face different offshoring cutoffs
  - Heterogeneous exposure: not everyone is equally at risk
- **Why does this apply to AI?**
  - AI = a technology that can perform certain tasks at low cost
  - Replace "offshoring cost" with "AI cost" — the mechanism is identical
  - Workers whose tasks AI can undercut face the same wage pressure as offshored workers

Model Setup

- **Tasks** indexed $i \in [0, 1]$ by AI susceptibility
  - Low i: routine cognitive (drafting, coding, data entry) — easy for AI
  - High i: complex, judgment-intensive — hard for AI
- **Production** via CES aggregator:

$$Y = \left[\int_0^1 y(i)^{\frac{\sigma-1}{\sigma}} \, di \right]^{\frac{\sigma}{\sigma-1}}$$

- **Workers:** type i^* specializes in task i^*; supply of task i is $f(i)$ (fixed)
- **AI cost** of task i: $c^{\mathrm{AI}}(i) = \tau \cdot e^{\alpha i}$
  - Lower τ = more capable AI
  - $e^{\alpha i}$ ensures complex tasks remain expensive for AI
- Competitive firms; output price P_Y fixed (small open economy)

Pre-AI Equilibrium: Derivation

Step 1 — Firm cost minimization gives task demand:

$$y^d(i) \;=\; Y \cdot \left(\frac{P_Y}{p_0(i)}\right)^{\sigma}$$

Higher task price $\rightarrow$ firm demands less of that task (standard substitution).

Step 2 — Market clearing $\big(y^d(i) = f(i)\big)$ pins down equilibrium task price:

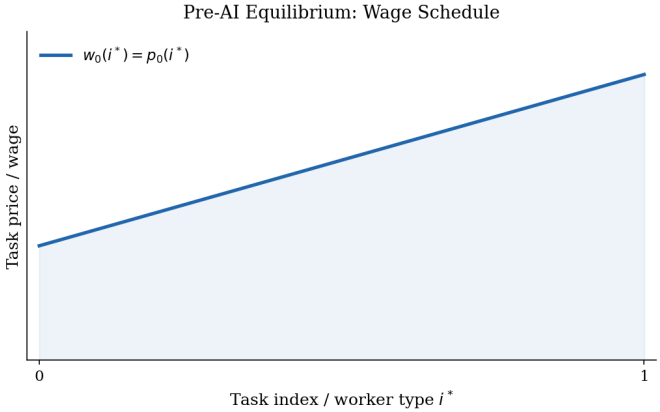
$$\boxed{p_0(i) \;=\; P_Y \cdot \left(\frac{Y}{f(i)}\right)^{1/\sigma}}$$

- Scarce worker types (low $f(i)$) command higher task prices
- Abundant worker types (high $f(i)$) earn less

Pre-AI Equilibrium: Wage Schedule

Step 3 — Under full specialization, each worker earns their task price:

$$w_0(i^*) \;=\; p_0(i^*)$$



- Each worker type i^* earns the competitive price of their task
- Higher-i^* workers earn more (their tasks are harder to replace)

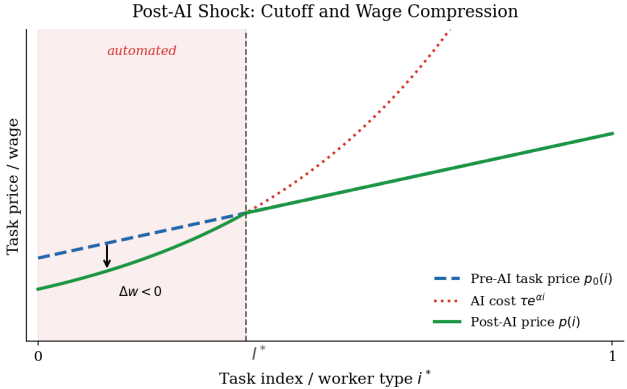
Note: upward slope is for illustration only — model imposes no shape restriction on p_0

Post-AI Shock: The Automation Cutoff

AI enters as a competing supplier. Competition drives task prices to:

$$p(i) = \min\big(p_0(i),\ \tau e^{\alpha i}\big)$$

A cutoff I^* emerges where AI cost equals pre-AI task price: $\tau e^{\alpha I^*} = p_0(I^*)$



- $i < I^*$: AI undercuts → wage compression

$$\Delta w(i^*) = \tau e^{\alpha i^*} - p_0(i^*) < 0$$

- $i \geq I^*$: human price prevails → unaffected

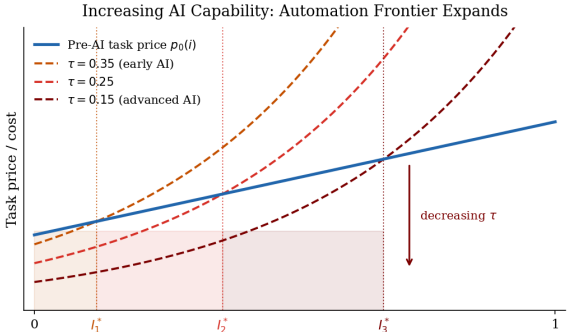
- Compressed wage < reservation wage → **exit employment**

What Determines the Cutoff I^*?

As AI improves ($\tau \downarrow$), I^* rises

$$\frac{\partial I^*}{\partial \tau} < 0$$

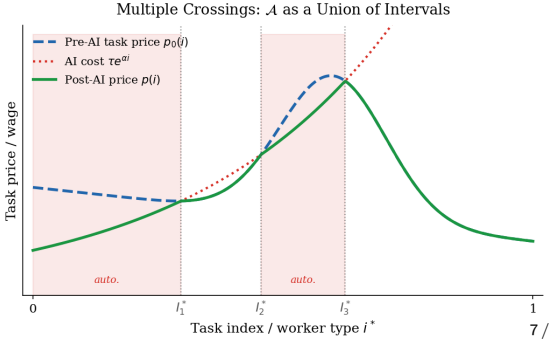
The automation frontier expands toward more complex tasks.



What if $p_0(i)$ is non-monotone?

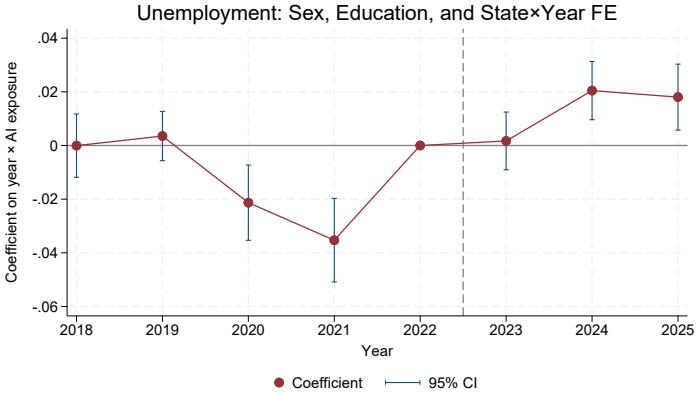
Cutoff may not be unique $\rightarrow$ automated set is a union of intervals:

$$\mathcal{A} = [0, I_1^*] \cup [I_2^*, I_3^*] \cup \cdots$$



Do We See This in the Data?

Unemployment × AI Exposure (CPS-ASEC, 2018–2025)



- Compare high vs. low AI-exposure workers, before vs. after ChatGPT (2022)
- Year 2022 = baseline
- **Post-2022:** high-exposure workers show rising unemployment
- Consistent with the model: workers below I^* face displacement

Thank you!

Questions welcome.

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